TELAMORPH



BUILT LIGHT. ENGINEERED STRONG.

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ENGINEERING-LED COMPOSITE MANUFACTURING

Telamorph is an engineering-led composite manufacturer based in Bulgaria. Founded by aeronautical engineers, we combine product engineering, tooling and in-house composite production to develop and manufacture lightweight structures for demanding applications.

WHAT WE DO

- **Geometry preparation** — Each project starts with defining a reliable 3D geometry, either from customer-supplied CAD data or developed through 3D scanning and reverse engineering when required.
- **Manufacturing review** — The geometry is then reviewed from a manufacturing perspective to define part splits, assemblies and the most efficient production approach, while preserving the customer's original design intent.
- **Laminate engineering** — Based on functional, structural and weight requirements, we define the composite laminate architecture, combining suitable fibers, resins, cores, coatings and local reinforcements.
- **Structural validation** — Where required, structural calculations and FEA are used to validate and optimize the proposed composite structure.
- **Process & tooling definition** — We then define the complete manufacturing route, including master-model and mould strategy, tooling materials, production method, curing requirements and post-processing.
- **First-article validation** — The selected tooling is manufactured and the first physical article is produced for dimensional, functional and structural validation.
- **Production readiness** — Any necessary adjustments are incorporated before the process is finalized and transferred into repeatable production.

WHERE WE ADD VALUE

- Turning CAD and engineering intent into repeatable composite production
- Reducing mass while maintaining stiffness, durability and finish requirements
- Delivering complex composite structures through one integrated manufacturing partner

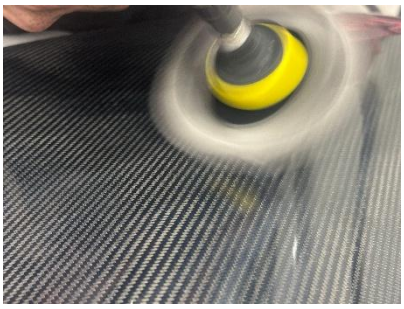
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SELECTED PROJECT 01

PORSCHE 911 - BESPOKE COMPOSITE BODYWORK

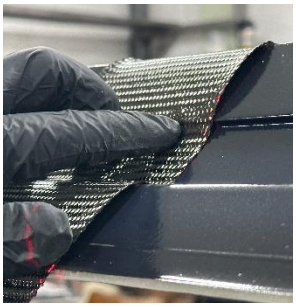
End-to-end engineering development and industrialisation of a complete composite body for the Porsche 911 (964), featuring extensively redesigned exterior geometry.











OUR RESPONSIBILITY

Starting from customer-supplied design data, Telamorph carried out the complete engineering development of the body, including all exterior panels and selected interior components. The scope covered material selection, structural design and sizing, component interfaces and joining solutions, body-to-chassis mounting systems, and design for manufacturability.

We subsequently designed and manufactured all required tooling and produced the first complete prototype body for validation against defined requirements for dimensional accuracy, fit, durability, reliability and structural performance. Following successful validation, the complete body was industrialised and introduced into low-volume production.

CONSTRUCTION

Hybrid carbon-fibre and carbon–Kevlar construction, optimised for manufacturability and high structural stiffness. Designed to withstand the defined service loads without permanent deformation while maintaining minimum structural weight.

RESULT

26 kg complete composite body, successfully integrated into approximately 15 vehicles. The project is fully documented and production-ready for a third manufacturing series.

MATERIAL	STRUCTURE	PROCESS	SCOPE
Hybrid	specialy chosen multylayer	VI / CNC trim / bond / fit / finishing	Complete vehicle

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SELECTED PROJECT 02

Monocoque cocpit and structure for UL Aircraft

Engineering development and manufacture of a composite monocoque structure for a certified aircraft.









OUR RESPONSIBILITY

Working closely with the customer's engineering team, We contributed to the aerodynamic geometry and developed the structural architecture, including load-bearing interfaces to the engine, propeller, main rotor and other highly loaded assemblies. The scope included structural sizing, laminate definition, thermal design around high-temperature zones, assembly design, tolerance definition and preparation for production.

We subsequently designed and manufactured the required tooling, performed dimensional control, and manufactured and assembled the first complete prototype structure for certification testing. Following successful validation and certification, the structure was industrialised and introduced into series production.

CONSTRUCTION

Each component uses a highly tailored composite architecture based on its specific load state. Fibre orientations and laminate thicknesses vary locally within individual parts, while stiffening closing panels limit deformation of lightly loaded aerodynamic surfaces. Honeycomb cores and different reinforcement fabrics are selected locally to achieve the required balance of stiffness, elasticity and weight.

RESULT

Successfully certified composite structure, subsequently introduced into series production and manufactured for nearly 70 aircraft. Several aircraft using the structure have now exceeded 2,500 flight hours and more that 5 years of real world testing.

MATERIAL	STRUCTURE	PROCESS	SCOPE
Carbon Fiber / Glass Fiber/ Cores	**Mass- Load optimized composite structure**	**Vacuum infusion / Prepreg / RTM**	**Complete structure**